

COMPASS Synoptic CB: Sulfide

July 2026 Samples

2026-07-28

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed
sample_month = "July"
sample_year = "2026"
Sample_Month_Year = "202607"
Run_Date = as.Date(c("07/20/2026", "07/24/2026", "07/28/2026"), format = "%m/%d/%Y")
Run_by = "Zoe Read, Melanie Giessner" #Instrument user
Script_run_by = "Zoe Read" #Code user

Run_notes="5/6 dups were good. 5/6 spikes were bad. Reran samples with high CVs." #any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE, pattern = "Plate")
(Raw_plates)

## [1] "Raw Data/COMPASS_H2S_20260720_Plate1_tidy.csv"
## [2] "Raw Data/COMPASS_H2S_20260720_Plate2_tidy.csv"
## [3] "Raw Data/COMPASS_H2S_20260720_Plate3_tidy.csv"
## [4] "Raw Data/COMPASS_H2S_20260724_Plate4_tidy.csv"
## [5] "Raw Data/COMPASS_H2S_20260724_Plate5_tidy.csv"
## [6] "Raw Data/COMPASS_H2S_20260724_Plate6_tidy.csv"
## [7] "Raw Data/COMPASS_H2S_20260728_Plate1_Reruns_tidy.csv"

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

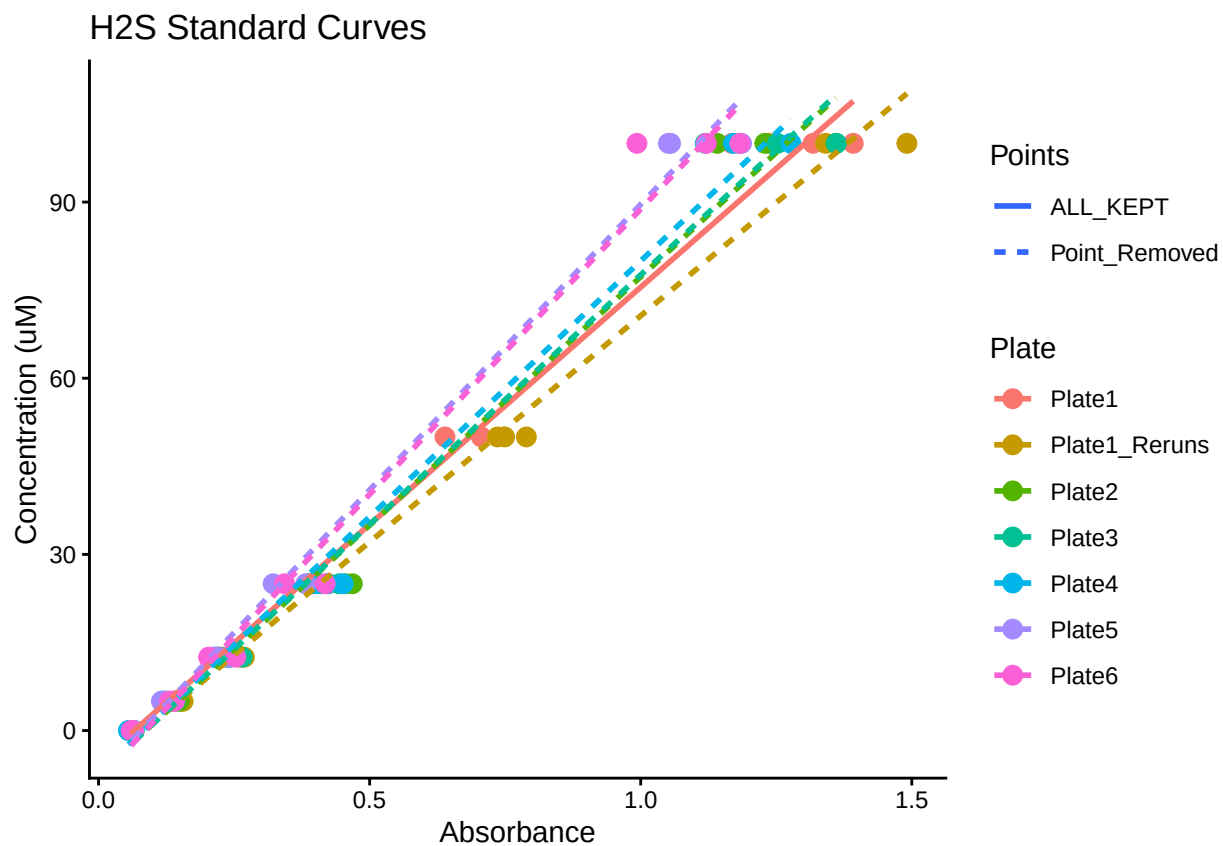
# Metadata path
Raw_Metadata <- "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_SynopticCB_H2S_202607_processed.csv"
rerun_path <- "Processed Data/COMPASS_SynopticCB_H2S_202607_reruns.csv"

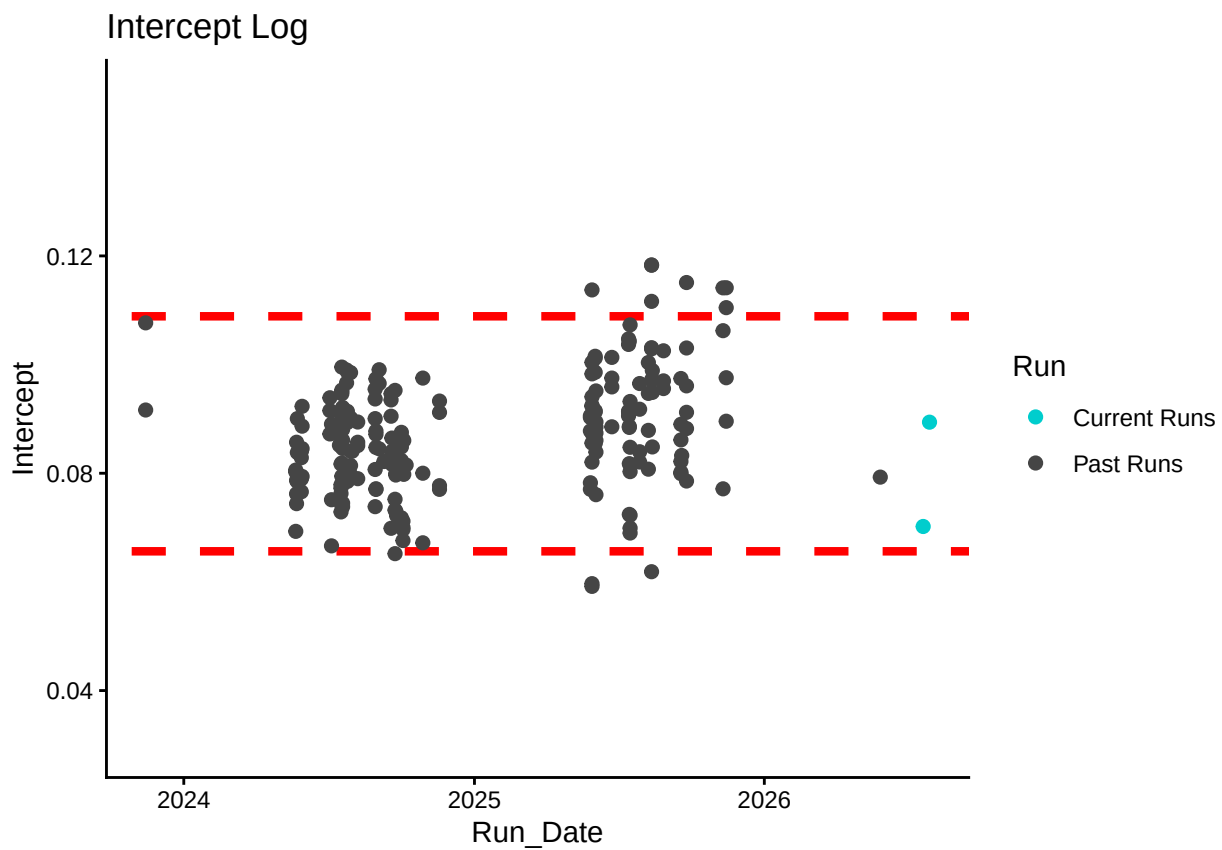
##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary
```

```
## Joining with 'by = join_by(Plate, IDs)'
```

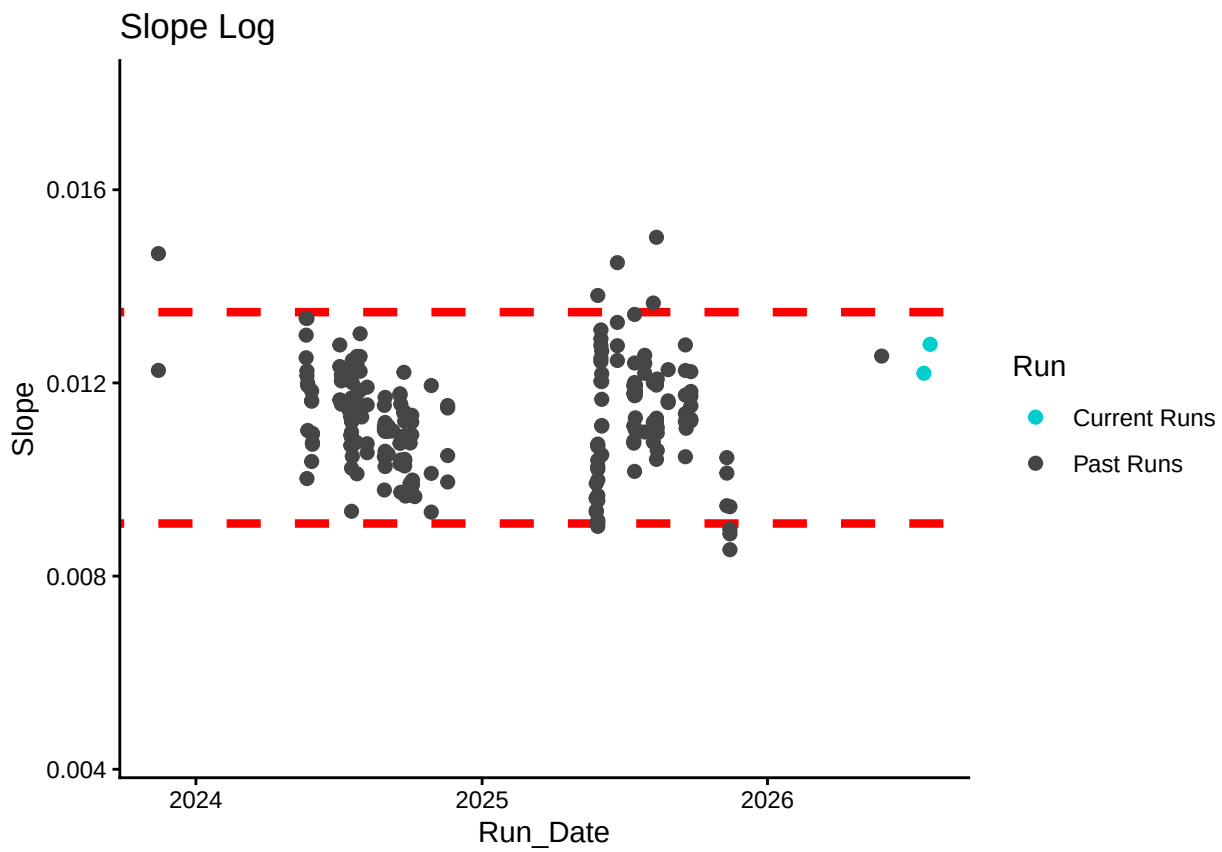
Plot Standard Curves



```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Warning: Removed 6 rows containing missing values or values outside the scale range
('geom_point()').

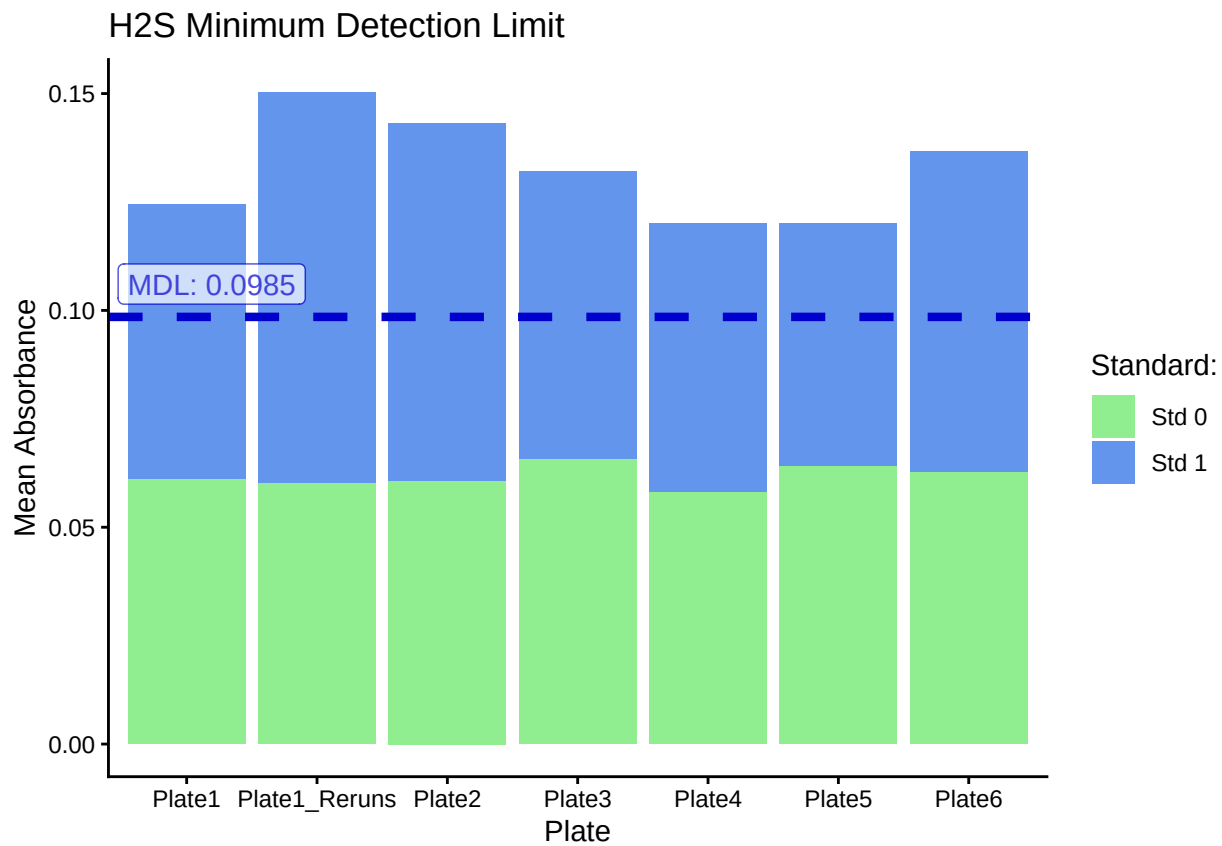


Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate1_Reruns	R2_Good	CV_Good
Plate2	R2_Bad	CV_Good
Plate3	R2_Bad	CV_Good
Plate4	R2_Good	CV_Good
Plate5	R2_Good	CV_Good
Plate5	R2_Good	CV_High
Plate6	R2_Bad	CV_Good
Plate6	R2_Bad	CV_High

Plate	Curve_RMVD
Plate1	KEPT
Plate1_Reruns	KEPT
Plate2	REMOVED
Plate3	REMOVED
Plate4	KEPT
Plate5	KEPT
Plate6	REMOVED

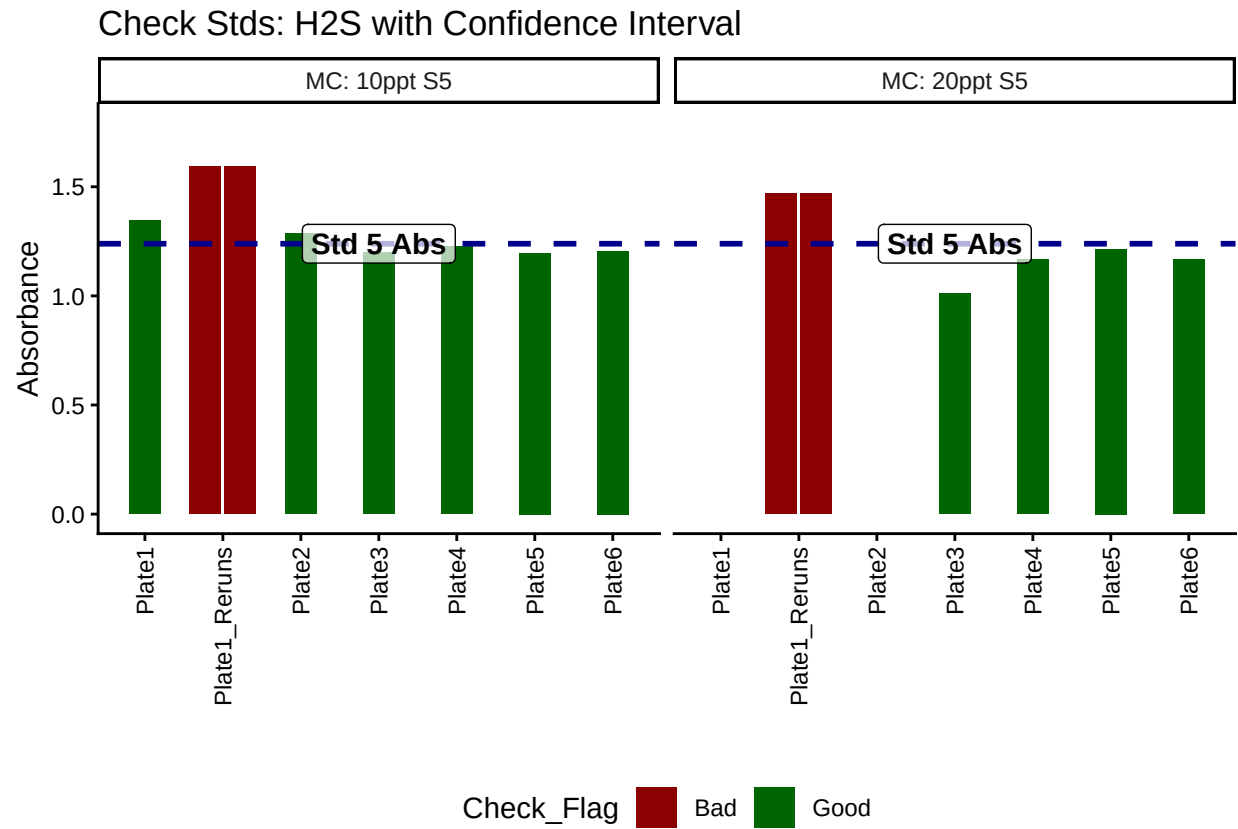
Method Detection Limit



Compare Check Standards to Standards

```
## Warning in left_join(H2S_Chks_bind, Chk_flags_plot, by = join_by(Std_ID, : Detected an unexpected many-
```

```
## i Row 11 of 'x' matches multiple rows in 'y'.
## i Row 3 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many"' to silence this warning.
```



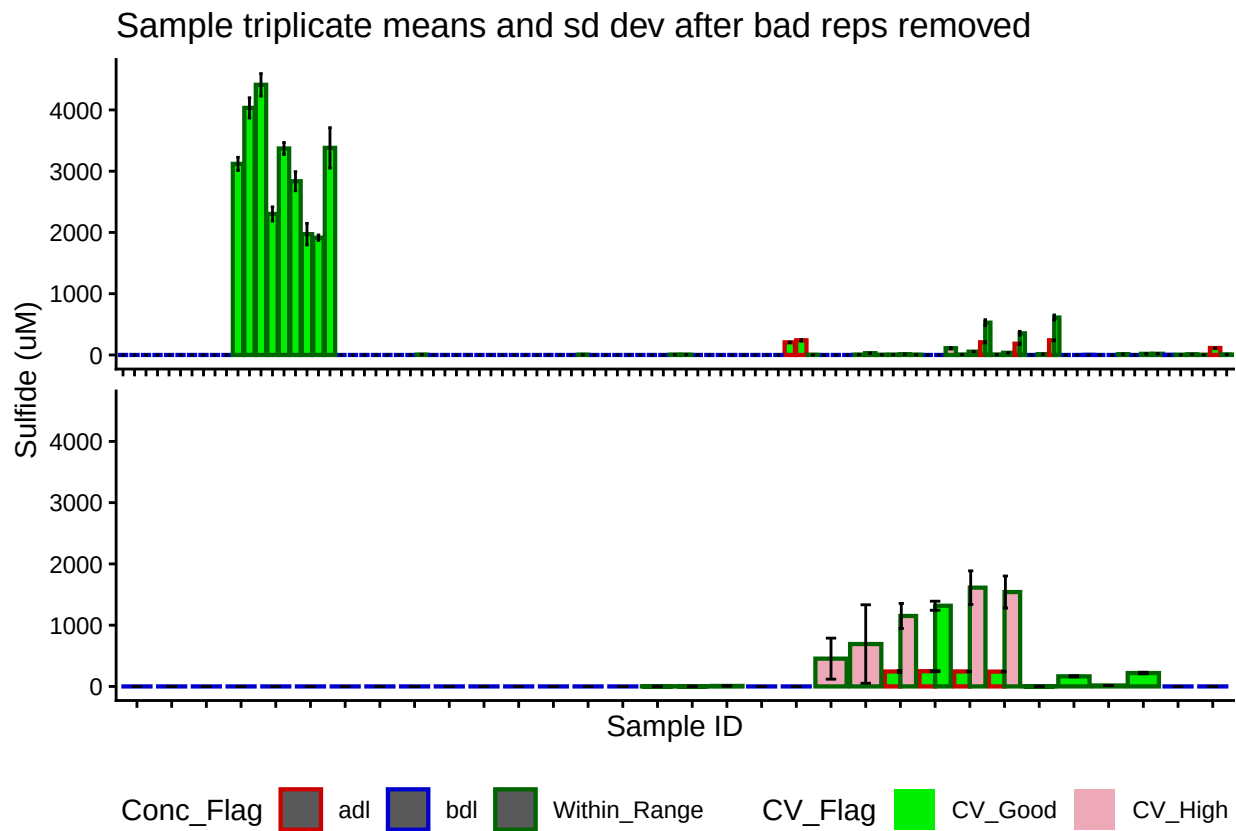
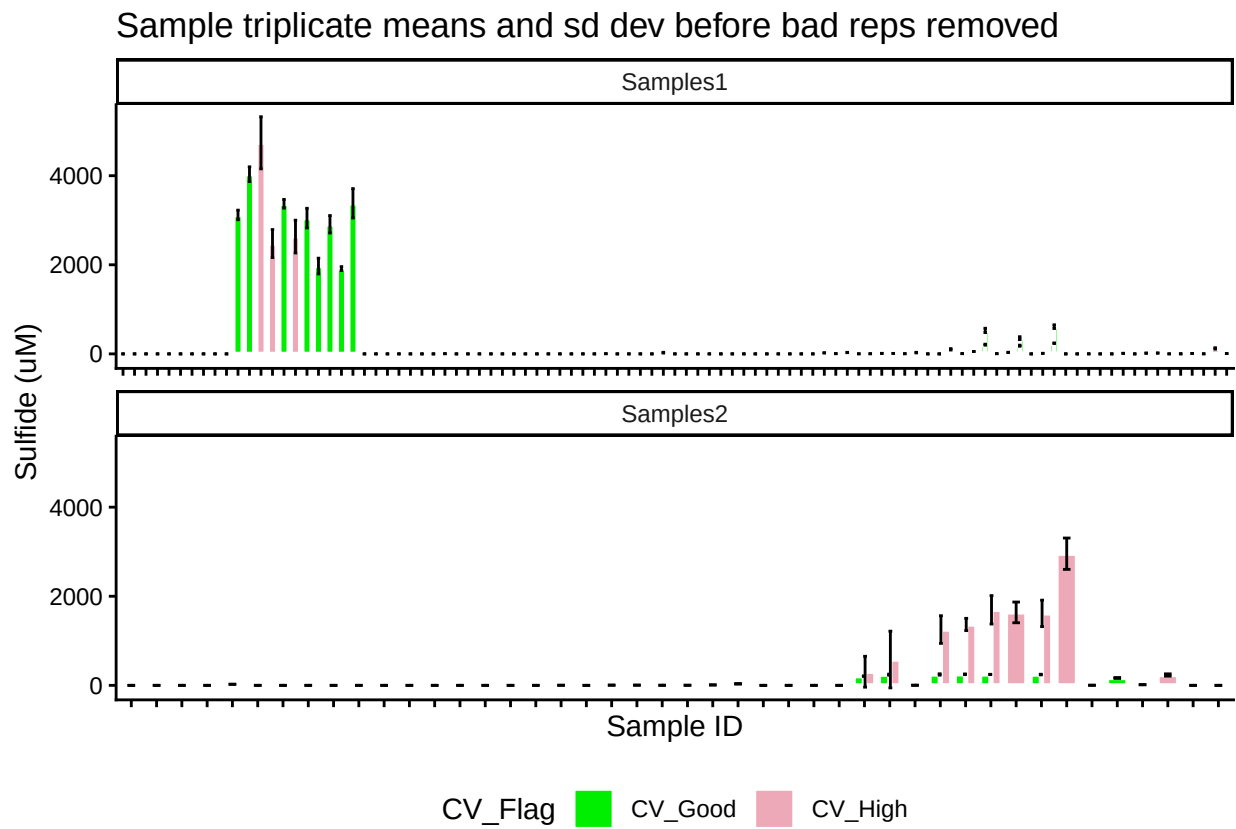
Display Any Check Flags

Plate	CHKs_Flag
Plate1_Reruns	MC: 10ppt S5 Chk_Bad
Plate1_Reruns	MC: 20ppt S5 Chk_Bad

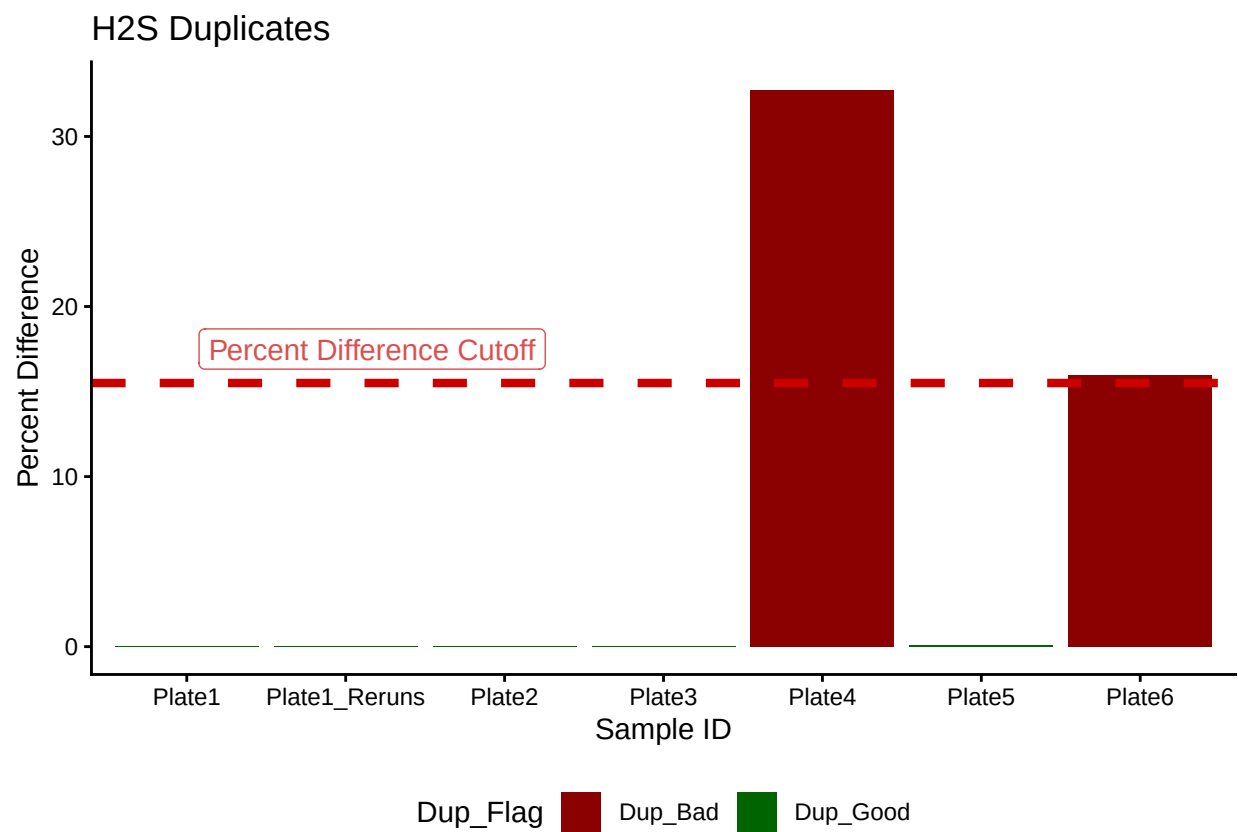
Plate	CHK_CV_Flags
-------	--------------

```
##Calculate Sulfide Concentrations & Add Concentration Flags
##Remove High CV Samples and Average
```

Plot Samples



Duplicate QAQC



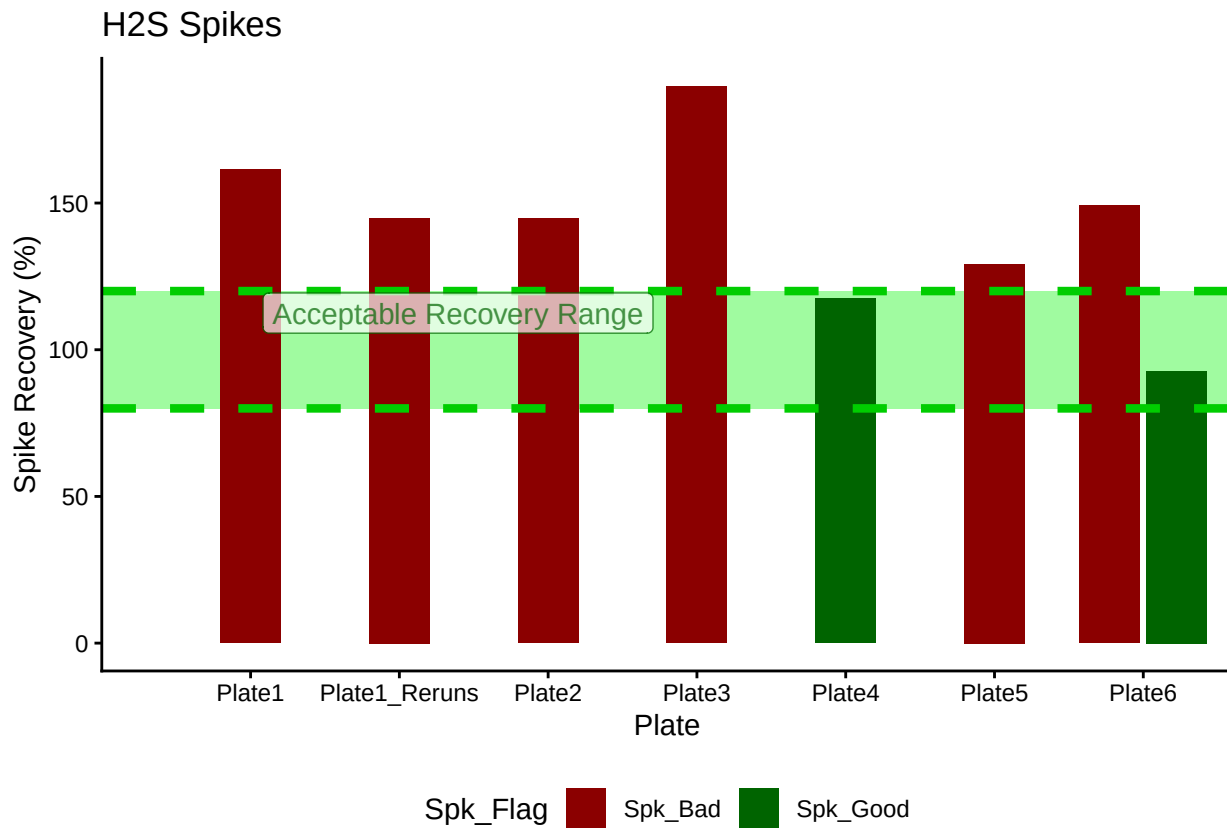
Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Good
Plate4	Dup_Bad
Plate5	Dup_Good
Plate6	Dup_Bad
Plate1_Reruns	Dup_Good

Spike QAQC

```
## Warning in left_join(., spk_flags_1, by = join_by(Spk_ID, Plate)): Detected an unexpected many-to-many :
## i Row 12 of 'x' matches multiple rows in 'y'.
## i Row 1 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many"' to silence this warning.
```



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad
Plate3	Spk_Bad
Plate4	Spk_Good
Plate5	Spk_Bad
Plate6	Spk_Bad
Plate1_Reruns	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

```
##Add Flags
```

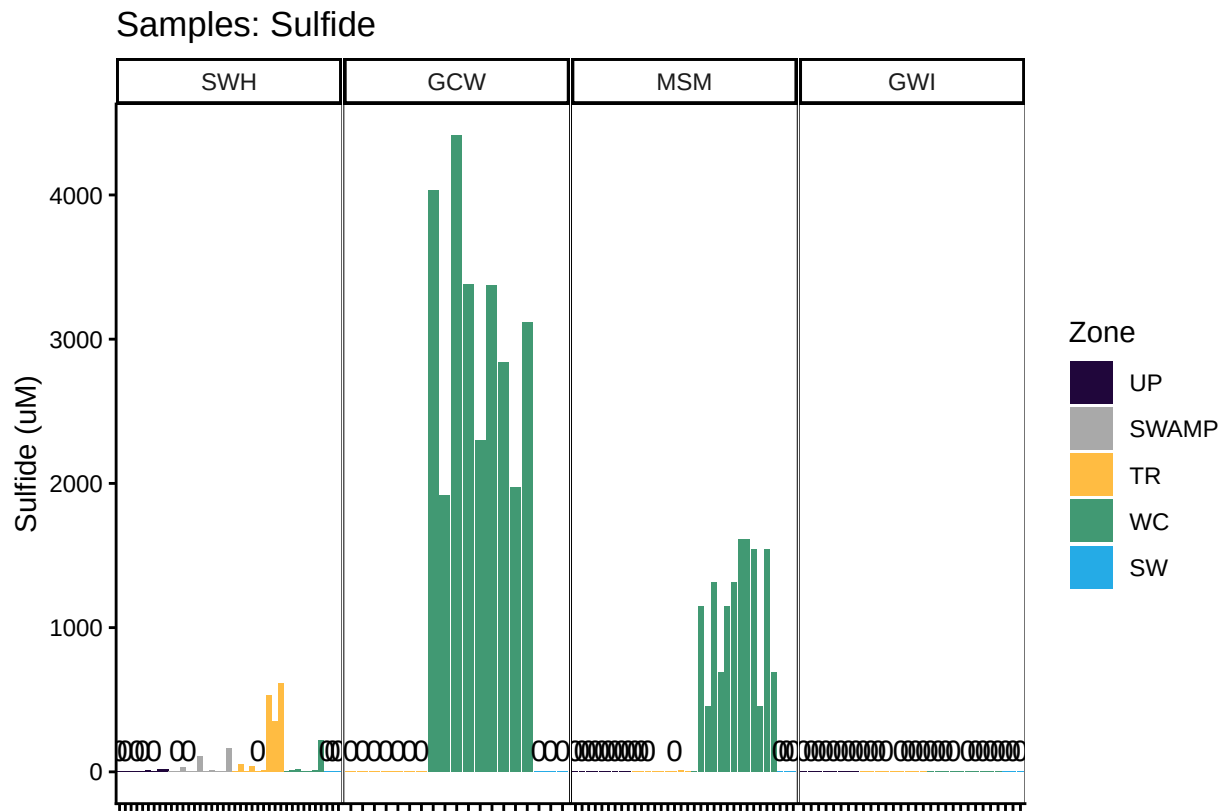
```
##Remove TEMPEST and FATE samples
```

```
Sample_data_IDs <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "TMP")) %>%
  filter(!str_detect(Sample_ID, "FATE")) %>%
  filter(!str_detect(Sample_ID, "dup")) %>%
  filter(!str_detect(Sample_ID, "spk"))
```


Check to see if samples run match metadata & merge info

All sample IDs are present in metadata.

Visualize Data by Plot



###END